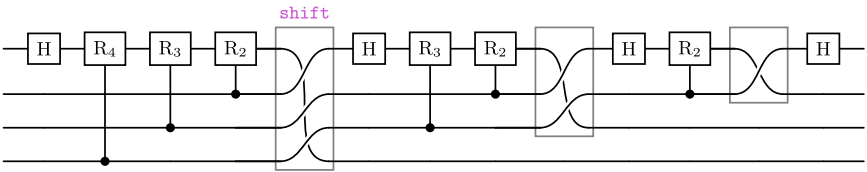```
1  decl qft(q̄){
2    q̄[1] *= H;
3    call rot(q̄);
4    call shift(q̄);
5    call qft(q̄ ⊖ [−1]); },
6  decl rot(q̄){
7    if |q̄| > 1 then
8      CPHASE(q̄[−1], q̄[1], |q̄|)
9      call rot(q̄ ⊖ [−1]);
10   else skip; },
11 decl shift(q̄){
12   if |q̄| > 1 then
13     SWAP(q̄[1], q̄[−1])
14     call shift(q̄ ⊖ [−1]);
15   else skip; }
16 ::
17 call qft(q̄);
```